Ten-Layer Analytic-Tail–Selector Frontier Review

Bin Wang

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Abstract

RH-252–RH-261 separate the deterministic target-tail problem from the legal-selector and quotient-tail problems. The deterministic numerator has an exact Cauchy all-order interface beyond the unit disk, but its boundary constant is not numerically certified. The order-28 anchor atlas and the margin-32 spectral window are finite diagnostics. The expanded single-use box, real conjugate-closed idempotent polynomial selectors supported on the resolved window, fractional signed fits, and unit-cap integer selectors are respectively obstructed within their stated scopes; the 23-endpoint quotient block-power computation remains finite and non-uniform. The updated gluing ledger has one of five obligations satisfied and zero complete certificates. Gates A–E remain explicitly false/open.

1 Scope and batch decision

The repository is the sole source for this review. The route coordinate inherited from RH-260 is

`legal_heads_obstructed_target_tail_exists_Ms_uncertified`
`quotient_finite_nonuniform_complete_certificate_zero`.

The review records 842 finite records under an explicit bookkeeping rule: 12 target-tail radius rows, 27 deterministic coefficient rows, 512 newly resolved roots, the endpoint rows of RH-254–RH-259, 64 RH-260 head-class endpoint cases, and 44 RH-260 consistency checks. The larger implicit class sizes are reported separately and are not promoted to an all-order result.

2 Exact interfaces

Let G be the deterministic one-step numerator from RH-46, let $r_H = 0.85$, and write $G_H(z) = G(z/r_H)$. The zero-free radius is $\rho = r_H\lambda = 1.42678748386407\ldots$, with $\lambda = 1.6785735104283177\ldots$.

Theorem 1 (Deterministic target tail)**.** *Assume G_H is holomorphic and nonzero on $|z| < \rho$, and choose the branch of $\log G_H$ at the origin. If*

$$\log G_H(z) = -\sum_{n\geq 1} a_n \frac{z^n}{n}, \qquad 0 \leq R < S < \rho,$$

then, with $M_S = \sup_{|z|=S} |\log G_H(z)| < \infty$,

$$\sum_{n\geq N} |a_n| \frac{R^n}{n} \leq M_S \frac{(R/S)^N}{1 - R/S}.$$

Proof. Cauchy's estimate for the coefficient of z^n in $\log G_H$ gives $|a_n|/n \leq M_S S^{-n}$. Summing the resulting geometric series for $n \geq N$ proves the claim. The statement is conditional only on the analytic branch and the choice $S < \rho$; RH-252 does not certify a numerical upper bound for M_S. $\square$

The finite-head/analytic-tail interface from RH-260 is equally precise. If H_N is the anchored finite-head logarithmic error, Q_N the quotient-tail budget, and A_N the target-tail budget, then

$$|L_N - L_*| \le H_N + Q_N + A_N, \qquad |e^{L_N} - e^{L_*}| \le e^{B_*}(e^{H_N+Q_N+A_N} - 1),$$

whenever $|L_*| \le B_*$. Uniform convergence requires all three budgets to vanish and also requires the coefficient bridge and a certified value of M_S. This is an interface theorem, not a present determinant certificate.

3 The ten layers

paper	layer	result	first missing theorem
RH-252	target tail	Exact analytic tail for $R < S < \rho$; M_S open	certified boundary supremum and cloud bridge
RH-253	anchor atlas	Exact deterministic dictionary through order 28	all-order cloud envelope; finite root fit is descriptive
RH-254	resolved window	32 endpoint atlases and 512 new roots; 21 complete, 11 split	legal anchored reachability
RH-255	expanded box	0/32 LP passes; binary class count 62,030,604,700	signed/complex or structurally new selector
RH-256	polynomial selector	Idempotents collapse to binary masks; real conjugate-closed resolved-window masks are obstructed	non-idempotent grouping, non-conjugate complex masks, or outside algebra
RH-257	signed moments	32/32 finite fits, all fractional; monodromy requires integer exponents	legal bounded integer selector
RH-258	unit-cap lattice	0/32 MILP passes; zero reported gap	larger cap with operator realization
RH-259	quotient power	23/23 finite C^{12} blocks contractive; worst $q_{12} = 0.505642$	uniform small-noise theorem and continuum bridge
RH-260	gluing ledger	Exact five-obligation ledger; one true component, zero complete certificates	legal head, bridge, uniform quotient tail, and certified M_S
RH-261	frontier review	Scoped synthesis and route firewall	a genuinely new legal head plus uniform tails

4 Finite findings and their boundaries

RH-253 extends the deterministic coefficient dictionary from orders 2–12 to 2–28. The new order-13–28 unit-disk logarithmic norm is 0.0021942543215719553, and a finite log-linear fit has root rate 0.7009986349256669. Neither number is an all-order theorem or a bridge to the moving noisy cloud [1, 2].

RH-254 resolves 16 additional roots at each of 32 endpoints. The maximum old/new matching discrepancy is $7.405469102694929 \times 10^{-9}$. The fixed-count boundary is material: 21 windows are shell-complete and 11 end in a split conjugate pair [3].

For the 32 shell-complete audits, RH-255 places every single-use shell choice in the convex box $0 \le w_j \le 1$. Its distance range is 0.14358493511963313–0.42399800369340307, with maximum LP gap $5.828670879282072 \times 10^{-15}$ and zero passes. This excludes the expanded box, its prefixes, and its binary subsets only; it does not exclude signed/complex selectors, larger windows, or operator constructions [4].

The algebraic selector firewall is exact. If $P = p(A)$ and $P^2 = P$, then on each generalized root space the restriction of P is either 0 or the identity. Complex polynomial coefficients are coordinates, not fractional multiplicities. The RH-255 box consequence is narrower: it excludes real, conjugate-closed idempotent masks supported on the resolved RH-254 roots, not a complex mask selecting only one member of a conjugate pair. Separately, the local moment product

$$\exp\left[-\sum_{n\geq 1}\left(\sum_j w_j\lambda_j^n\right)\frac{z^n}{n}\right]=\prod_j(1-\lambda_j z)^{w_j}$$

is single-valued meromorphic around each reciprocal root only when combined weights are integral. Thus the 32/32 arbitrary signed fits in RH-257 are finite interpolation facts with nontrivial monodromy, not legal determinant selectors [5, 6].

RH-258 audits the first monodromy-legal lattice $w_j \in \{-1, 0, 1\}$. All 32 MILPs have zero reported gap and no pass; distances are 0.10607370900129424–0.3534900682213731. The implicit aggregate search contains 39,417,456,084,975,216 lattice points. This is a unit-cap negative result, not a larger-cap or operator-realization theorem [7].

RH-259 extends the ordered-Schur quotient computation to dimension 1024. All 23 audited twelfth powers are contractive, but q_{12} ranges from 0.22185212659640824 to 0.5056418005507071, a deterioration factor of 1.2856404093172868 relative to RH-246. The finite unit-disk tail diagnostic is 0.0005654507945432548; it is floating-point evidence, not an interval enclosure, and nine archived endpoints plus the continuum bridge are open [8].

5 Updated gluing ledger

The five logically independent components are

$$(\text{legal head},\ \text{coefficient bridge},\ \text{uniform quotient tail},\ \text{analytic target tail},\ \text{certified } M_S).$$

RH-260 archives the vector

$$(\text{false}, \text{false}, \text{false}, \text{true}, \text{false}),$$

so exactly one component is satisfied and the complete-certificate count is zero. The head obstruction is explicitly limited to the two audited expanded classes (64 endpoint/class cases); it is not a claim that every possible selector fails. The target-tail theorem remains useful because it isolates the missing numerical M_S certificate from the independent cloud bridge [9].

6 Gate audit and next triggers

gate	requirement	status after RH-261
A	canonical intrinsic dynamical determinant and identification	false/open
B	time-oriented scattering or unitary completion	false/open
C	self-adjoint generator and intrinsic $T \log T$ law	false/open
D	von Mangoldt-weighted prime-power traces	false/open
E	equality with the completed-zeta divisor	false/open

The next admissible reopening inputs are: (i) a rigorous computable M_S bound; (ii) a legal invariant selector outside the audited classes together with a coefficient bridge; (iii) a uniform quotient block-power theorem with the missing endpoint and continuum bridge; or (iv) a larger integer-cap audit paired with an actual operator realization. Repeating finite fits, frozen box reweighting, or isolated Schur computations does not meet these triggers.

No section constructs a Hilbert–Polya operator, identifies Riemann zeros, proves a zeta-divisor equality, or implies RH.

References

[1] Bin Wang. Deterministic-numerator analytic-tail certificate, 2026. RH-252 repository paper.

[2] Bin Wang. Extended deterministic anchor atlas, 2026. RH-253 repository paper.

[3] Bin Wang. Expanded resolved candidate-window atlas, 2026. RH-254 repository paper.

[4] Bin Wang. Expanded-window anchored zonotope obstruction, 2026. RH-255 repository paper.

[5] Bin Wang. Invariant polynomial-selector binary collapse, 2026. RH-256 repository paper.

[6] Bin Wang. Monodromy integrality barrier for signed moment fits, 2026. RH-257 repository paper.

[7] Bin Wang. Unit-cap signed-integer selector obstruction, 2026. RH-258 repository paper.

[8] Bin Wang. Extended quotient block-power diagnostic, 2026. RH-259 repository paper.

[9] Bin Wang. Updated anchored head-tail certificate ledger, 2026. RH-260 repository paper.